MILITARY SPECIFICATION 9 February 1978

# RECEIVER-TRANSMITTER RT-1185( )/GRA-114

This specification is approved for use by Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the Receiver-Transmitter RT-1185( )/GRA-114 which is used by the sound observer as part of Radio Data Link, Sound Ranging AN/GRA-114 (See 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| NN-P-71 | Pallets, Materials-handling (General Construction Requirements) |
| QQ-S-781 | Strapping, Steel, Flat and Seals |
| PPP-B-585 | Box, Wood, Wirebound |
| PPP-B-601 | Box, Wood, Cleated-Plywood |
| PPP-B-621 | Box, Wood, Nailed and Lock-Corner |
| PPP-B-636 | Box, Fiberboard |
| PPP-B-640 | Box, Fiberboard, Corrugated, Triple-wall |
| PPP-F-320 | Fiberboard, Corrugated and Solid, Sheet Stock Container Grade and Cut Shapes |
| PPP-S-760 | Strapping, Nonmetallic (and Connectors) |
| PPP-T-97 | Tape, Pressure-Sensitive Adhesive, Filament Reinforced |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronic Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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| MIL-P-116 | Preservation-packaging, Methods of |
| MIL-S-3950 | Switches, Toggle, General Specification for |
| MIL-P-11268 | Parts, Materials and Processes Used in Electronic Equipment |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finishes for Ground Electronic Equipment |
| MIL-P-19644 | Plastic Molding Material (Polystyrene Foam, Expanded Bead) |
| MIL-P-26514 | Polyurethane Foam, Rigid or Plastic, for Packaging |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-356 | Commercial Packaging of Supplies and Equipment |

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| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspections by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-147 | Palletized and Containerized Unit Loads 40 Inch X 48 Inch Pallets, Skids, Runners or Pallet Type Base |
| MIL-STD-252 | Wired Equipment, Classification of Visual and Mechanical Defects |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-461 | Electromagnetic Interference Characteristics, Requirements for Equipment |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement of |
| MIL-STD-781 | Reliability Tests: Exponential Distribution |
| MIL-STD-810 | Environmental Test Methods |

DRAWINGS

ELECTRONICS COMMAND

| | |
|---|---|
| SC-A-678034 | Burn-in Cycle |
| SM-A-865231 | Performance Test Procedures |



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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3 (see 6.2(g)).

3.2 Construction. The Receiver-Transmitter RT-1185( )/GRA-114 shall be constructed in accordance with the requirements of this specification and the model.

3.2.1 Knobs. Knobs on code switches shall be provided with key-ways that insure proper orientation on the switch shafts.

3.2.2 Printed circuit boards. Where feasible, printed circuit boards shall be provided with pull-out flanges for easy removal. Flanges shall be labeled.

3.2.3 Test points. Test points shall be provided on all of the printed circuit boards. Test points shall be properly labeled and located so as to facilitate testing without removal of the boards from the chassis wherever feasible.

3.2.4 Panel. The control panel shall be laid out in a manner which permits ready access and manipulation of controls and connectors by personnel wearing arctic gloves.

3.2.5 Encode reliability. The RT-1185 shall provide a command signal with a single actuation of the Data Start button.

3.2.6 Gas emission. Electrical arcing shall be prevented in the electronic compartments of RT-1183. Battery compartment shall be isolated from the electronic compartment by means of air-tight and chemical tight bulkheads.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply.



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3.3.1 Toggle switches. Toggle switches, where employed, shall be full size in accordance with MIL-S-3950.

3.4 Finish. All surfaces of the RT-1185 requiring a protective coating shall be finished in accordance with MIL-F-14072 (see 4.4).

3.5 Marking.

3.5.1 General. Marking shall be in accordance with MIL-M-13231 (see 4.4). Front panel markings shall be group I as described in that specification. Shield cans shall be clearly marked to indicate the character and relative position of their interior contents.

3.5.2 Visibility. Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. Each RT-1185 shall be serial-numbered (see 4.4).

3.6 Interchangeability. Like units, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.6).

3.7 Burn-in. The RT-1185 shall meet the requirements of this specification after being subjected to the burn-in cycle of Drawing SC-A-678034 (see 4.7).

3.8 Performance requirements.

3.8.1 Transmitter.

3.8.1.1 Power output. The output power of each transmitter channel of RT-1185 shall meet the requirements of Table I (see 4.9).

TABLE I. Power output

| Input (voltage) | Setting | Power Output |
|---|---|---|
| 10.0 | HI | Not less than 3.57 watts |
| 12.0 | HI | Not less than 5.0 watts |
| 15.0 | HI | Not less than 5.0 watts |
| 12.0 | MED | Not less than 250 mw and not more than 1000 mw |



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3.8.1.2 Efficiency. The efficiency shall be at least 32 per cent (see 4.9).

3.8.1.3 Encode operation. When the command switch is depressed, an appropriate set of code groups shall be transmitted (see 4.9).

3.8.1.4 Remote encode control. With a power supply connected to the remote terminals, switch on the power supply. Command transmissions shall be initiated with a positive voltage of 4 volts or more (see 4.9).

3.8.1.5 Code channel frequency accuracy. The channel frequency shall be within ±.007 percent of the assigned frequency (see 4.9).

3.8.1.6 Voice channel frequency accuracy. The channel frequency shall be within ±.007 percent of the assigned frequency (see 4.9).

3.8.1.7 Deviation limiting. The deviation shall not be greater than ±19.0 KHz when a 1.0 KHz signal of 10 mv is applied to the input of the handset connector (see 4.9).

3.8.1.8 Local voice modulation capability. The FM deviation shall not be less than 7 KHz nor more than 10 KHz when a 1.0 KHz signal of 1 mv is applied to the input of the handset connector (see 4.9).

3.8.1.9 Local voice distortion. The voice distortion shall be less than 5 percent with a carrier deviation of 7 to 10 KHz at a frequency of 1000 Hz (see 4.9).

3.8.1.10 Local voice frequency response. The voice frequency response shall meet the requirements of Table II (see 4.9).

TABLE II. Local voice frequency response.

| Frequency (Hz) | Response (dB) |
|---|---|
| 150 | -3.0 +2.0 |
| 300 | -1.0 ±2.0 |
| 600 | 0 ±1.0 |
| 1000 | 0 reference |
| 3500 | -3.0 +2.0 |
| 5000 | -6.0 min |



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3.8.1.11 Remote voice control operation. An R.F carrier shall be present with a negative voltage of between 1 and 4 volts (see 4.9).

3.8.1.12 Remote voice modulation gains. The FM deviation shall not be less than 7 KHz nor more than 10 KHz when a 1.0 KHz signal of 0.8 VRMS is applied at the input to the remote terminals (see 4.9).

3.8.1.13 Transmit remote voice distortion. The voice distortion shall be less than 5 percent with a carrier deviation of 7 to 10 KHz at a frequency of 1000 Hz (see 4.9).

3.8.1.14 Remote voice frequency response. The voice frequency response shall meet the requirements of Table III (see 4.9).

TABLE III. Remote voice frequency response.

| Frequency (Hz) | Response (dB) |
|---|---|
| 150 | -1.0 ±1.0 |
| 300 | 0 ± .5 |
| 600 | 0 ± .5 |
| 1000 | 0 reference |
| 3500 | -3.0 +2.0 |
| 5000 | -6.0 dB min |

3.8.2 Receiver.

3.8.2.1 Sensitivity. The sensitivity shall be such that 0.5 uv applied to the input terminals of the receiver shall produce a signal-plus-noise-to-noise ratio of at least 10 dB (see 4.9).

3.8.2.2 Limiting. Not more than 1.0 uv of input signal shall be required to achieve full limiting. The limiting characteristics shall be such that the audio power output shall not vary more than ±3 dB at RF input levels of 1 to 100,000 uv (see 4.9).

3.8.2.3 Audio output power. The output power measured at the handset jack across 1000 ohms shall not be less than 5 mw at 12 volts battery supply and 3 mw at 10 volts supply. Distortion shall not exceed 5% (see 4.9).



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3.8.2.4 Remote voice gain. An RF carrier of the proper frequency which is deviated + 8 KHz by a one KHz audio signal shall produce an output signal at the remote terminals of 0.8 VRMS ± 20 percent (see 4.9).

3.8.2.5 Receive remote voice distortion. The distortion in audio output signals at the remote terminals shall not exceed 5 per cent (see 4.9).

3.8.2.6 Remote voice response. The frequency response shall be in accordance with Table IV (see 4.9).

TABLE IV. Remote voice response.

| Frequency (Hz) | Response (dB) |
|---|---|
| 150 | -3.0 ±1.5 |
| 300 | -1.0 ±1.0 |
| 600 | 0 ±1.0 |
| 1000 | 0 Reference |
| 3500 | -3.0 ±2.0 |
| 5000 | -5.0 min |

3.8.2.7 Remote line bias The voltage at the remote terminals shall be 1.0 Vdc ±10 percent (see 4.9).

3.8.2.8 Standby input power. The standby current supply shall not exceed 22 ma (see 4.9).

3.8.2.9 Front End Overload. An interfering signal 120 dB above 1 microvolt and removed ± 15 Mhz from the desired frequency shall not cause failure of any part of RT-1185.

3.9 Service conditions. The RT-1185 shall meet the service conditions below.

3.9.1 Temperature.

3.9.1.1 High temperature. During testing as specified in 4.10.1.1, the RT-1185 shall meet the following requirements:

| Paragraph | Measurement |
|---|---|
| 3.8.1.1 | Power output |
| 3.8.1.3 | Encode operation |
| 3.8.1.4 | Remote encode control |
| 3.8.1.5 | Code channel frequency accuracy |



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| Paragraph | Measurement |
|---|---|
| 3.8.1.7 | Deviation limiting |
| 3.8.1.8 | Local voice modulation capability |
| 3.8.1.9 | Local voice distortion |
| 3.8.1.11 | Remote voice control operation |
| 3.8.1.12 | Remote voice modulation gains |
| 3.8.1.13 | Transmit remote voice distortion |
| 3.8.1.14 | Remote voice frequency response |
| 3.8.2.1 | Sensitivity |
| 3.8.2.2 | Limiting |
| 3.8.2.3 | Audio output power |

3.9.1.2 Low temperature. During testing as specified in 4.10.1.2, the RT-1185 shall meet the following requirements:

| Paragraph | Measurement |
|---|---|
| 3.8.1.1 | Power output |
| 3.8.1.3 | Encode operation |
| 3.8.1.4 | Remote encode control |
| 3.8.1.5 | Code channel frequency accuracy |
| 3.8.1.7 | Deviation limiting |
| 3.8.1.8 | Local voice modulation capability |
| 3.8.1.9 | Local voice distortion |
| 3.8.1.11 | Remote voice control operation |
| 3.8.1.12 | Remote voice modulation gains |
| 3.8.1.13 | Transmit remote voice distortion |
| 3.8.1.14 | Remote voice frequency response |
| 3.8.2.1 | Sensitivity |
| 3.8.2.2 | Limiting |
| 3.8.2.3 | Audio output power |

3.9.2 Humidity. The RT-1185 shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the following requirements during the operating times specified in 4.10.2.

| Paragraph | Measurement |
|---|---|
| 3.8.1.1 | Power output |
| 3.8.1.3 | Encode operation |
| 3.8.1.4 | Remote encode control |
| 3.8.1.5 | Code channel frequency accuracy |



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| Paragraph | Measurement |
|---|---|
| 3.8.1.7 | Deviation limiting |
| 3.8.1.8 | Local voice modulation capability |
| 3.8.1.9 | Local voice distortion |
| 3.8.1.11 | Remote voice control operation |
| 3.8.1.12 | Remote voice modulation gains |
| 3.8.1.13 | Transmit remote voice distortion |
| 3.8.1.14 | Remote voice frequency response |
| 3.8.2.1 | Sensitivity |
| 3.8.2.2 | Limiting |
| 3.8.2.3 | Audio output power |

3.9.3 Immersion. The RT-1185 shall show no evidence of leakage after it is tested as specified in 4.10.3.

3.9.4 Dust. After testing as specified in 4.10.4, the RT-1185 shall meet full specification performance for the following measurements:

| Paragraph | Measurement |
|---|---|
| 3.8.1.1 | Power output |
| 3.8.2.1 | Sensitivity |
| 3.8.1.9. | Local voice distortion |

In addition all top panel controls shall have unrestricted freedom of motion and all controls and fittings shall be capable of performing their intended function.

3.9.5 Vibration, bounce and shock.

3.9.5.1 Bounce and shock. The RT-1185 shall meet the requirements of Table V.

TABLE V. Bounce and shock

| Paragraph heading | Test Para | Performance after test |
|---|---|---|
| Bounce | 4.10.5.1 | Specified performance. No physical damage except minor surface abrasion. 1/ |
| Shock; drop | 4.10.5.2 | Specified performance. Any physical damage shall be minor only. 1/ |



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1/ The units shall meet specified performance for the following:

| Paragraph | Measurement |
|---|---|
| 3.8.1.1 | Power output |
| 3.8.1.3 | Encode operation |
| 3.8.1.4 | Remote encode control |
| 3.8.1.5 | Code channel frequency accuracy |
| 3.8.1.6 | Voice channel frequency accuracy |
| 3.8.1.7 | Deviation limiting |
| 3.8.1.8 | Local voice modulation capability |
| 3.8.1.9 | Local voice distortion |
| 3.8.1.11 | Remote voice control operation |
| 3.8.1.12 | Remote voice modulation gains |
| 3.8.1.13 | Transmit remote voice distortion |
| 3.8.1.14 | Remote voice frequency response |
| 3.8.2.1 | Sensitivity |
| 3.8.2.2 | Limiting |
| 3.8.2.3 | Audio output power |

3.9.6 Salt fog. After the salt fog test of 4.10.6, the RT-1185 when examined visually, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface. Electrical operation is not required.

3.9.7 Fungus. After the fungus test of 4.10.7 the RT-1185 shall be visually examined and shall reveal no evidence of viable fungus on the surface.

3.10 Electromagnetic interference characteristics. The RT-1185 shall meet the emission and susceptibility requirements for MIL-STD-461 for Class I equipments (see 4.11). Test set-ups and test procedure shall conform to the measurement techniques of MIL-STD-462. The following requirements are applicable:

| Transmit Mode | Receive Mode |
|---|---|
| CE06 3/ | CS03 CE06 |
| RE02 1/ | CS04 2/ RE02 1/ |
| RS03 1/ | CS05 RE03 1/ |

1/ The upper frequency of the test shall be 1000 MHz. In the Transmit mode "on channel" radiations from the RT-1185 are not considered to be spurious signals.

2/ All responses shall be 96 dB down except for a maximum of five (5) responses which shall not be less than 80 dB down, when tested in accordance with MIL-STD-462.



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3/ a. The equipment shall meet the spurious and harmonic output requirements of MIL-STD-461A, Notice 4, except for a total of three responses which may be between 70 and 80dB down. No response will be less than 70dB down relative to the fundamental power in the medium power mode.

b. The equipment shall meet the spurious and harmonic output requirements of MIL-STD-461A, Notice 4, except for a total of three responses which may be between 70 and 80dB down. No response will be less than 70dB down. The limit at the second harmonic shall be relaxed by 5dB when operated in the high power mode.

3.11 Weight. The RT-1185 shall weigh no more than 11.5 pounds, less batteries and handset (see 4.12).

3.12 Reliability. The RT-1185( )/GRA-114 shall have a specified MTBF ($\theta^{o}$) of 1040 hours. The test specified in paragraph 4.14 shall be conducted to determine compliance with this reliability requirement.

3.13 Workmanship. The RT-1185 shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268 (see 4.4):

General requirements for plastic material and plastic parts

Wiring and cabling, including:

Slack
Protection
Clearance
Splicing
Connections, general
Grounding, general
Shielding on wire and cable
Soldering and brazing
Cleaning of units
General requirements for securing of parts
Cleaning prior to welding
Controls
Self tapping screws
Cleaning after welding
Printed wiring assemblies



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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). Does not include packaging.

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.8).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.



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4.3.1 First article units. The first article inspection shall consist of the inspections specified in Table VI,. and shall be performed in the order specified. Seven units shall be provided for Group A,B,C, and D inspections.

4.3.2 First article data. The first article test plan and test report shall be as required in the contract or purchase order.

TABLE VI. First article inspection.

| | Reqt Para | Test Para | Unit 1 | Unit 2 | Unit 3 | Unit 4 | Unit 5 | Units 6-7 |
|---|---|---|---|---|---|---|---|---|
| 1. Inspections covered subsidiary documents. 1/ | 3.3<br>3.4<br>3.5<br>3.13 | 4.4 | Inspection to be performed on all units. | | | | | |
| 2. Burn-in 1/ | 3.7 | 4.7 | Inspection to be performed on all units. | | | | | |
| 3. Group A inspection 1/ See Table VII | | | Inspection to be performed on all units. | | | | | |
| 4. Group B inspection 1/ See Table VIII | | | Inspection to be performed on all units. | | | | | |
| 5. Group C inspection 2/ | | | | | | | | |
| Receiver input Power | 3.8.2.8 | 4.9 | | | 1 | | | |
| High temperature | 3.9.1.1 | 4.10.1.1 | | 1 | | | | |
| Low temperature | 3.9.1.2 | 4.10.1.2 | | 2 | | | | |
| Humidity | 3.9.2 | 4.10.2 | | 3 | | | | |
| Immersion | 3.9.3 | 4.10.3 | 3 | | | | | |
| Dust | 3.9.4 | 4.10.4 | | | 2 | | | |
| Bounce | 3.9.5.1 | 4.10.5.1 | | | 3 | | | |
| Shock; drop | 3.9.5.1 | 4.10.5.2 | 4 | | | | | |
| Salt fog | 3.9.6 | 4.10.6 | | | | 1 | | |
| Fungus | 3.9.7 | 4.10.7 | | | | | 1 | |
| EMI | 3.10 | 4.11 | 2 | | | | | |
| Weight | 3.11 | 4.12 | 1 | | | | | |
| 6. Group D inspection | 3.12 | 4.14 | | | | | | 1 |



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1/ The inspection 1 to 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

2/ The numbers in the unit columns in the table, for inspection 5 specify the order of testing for the indicated unit.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

| Item | Where required |
|---|---|
| Parts, material, and processes; general | 3.3 |
| Finish | 3.4 |
| Marking | 3.5 |
| Workmanship | 3.13 |

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1, 4.5.2, 4.5.3, 4.5.4 and 4.7. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table VII. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table VII.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.



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TABLE VII. Group A inspection.

| Inspection | Reqt. Para. | Test Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Visual and mechanical | 3.13 | 4.4 | 1% | 4% |
| Electrical | | | | 1/ |
| Power output | 3.8.1.1 | 4.9 | 1% | |
| Encode operation | 3.8.1.3 | 4.9 | 1% | |
| Remote encode control | 3.8.1.4 | 4.9 | 1% | |
| Code channel frequency accuracy | 3.8.1.5 | 4.9 | 1% | |
| Local voice distortion | 3.8.1.9 | 4.9 | 1% | |
| Remote voice control operation | 3.8.1.11 | 4.9 | 1% | |
| Remote voice modulation gains | 3.8.1.12 | 4.9 | 1% | |
| Sensitivity | 3.8.2.1 | 4.9 | 1% | |
| Limiting | 3.8.2.2 | 4.9 | 1% | |
| Audio output power | 3.8.2.3 | 4.9 | 1% | |
| Remote voice gain | 3.8.2.4 | 4.9 | 1% | |

1/ All electrical defects are major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table VIII and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sample plans, for the AQL's listed in Table VIII, shall be as follows:

| AQL | Inspection level |
|---|---|
| 4% | S-4 |
| 6.5% | S-4 |

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.



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TABLE VIII. Group B inspection.

| Inspection | Requirement Paragraph | Test Paragraph | AQL |
|---|---|---|---|
| Interchangeability | 3.6 | 4.6 | 6.5% |
| Electrical | | | |
| Efficiency | 3.8.1.2 | 4.9 | 4.0% |
| Voice channel frequency accuracy | 3.8.1.6 | 4.9 | 4.0% |
| Deviation limiting | 3.8.1.7 | 4.9 | 4.0% |
| Local voice modulation capability | 3.8.1.8 | 4.9 | 4.0% |
| Local voice frequency response | 3.8.1.10 | 4.9 | 4.0% |
| Transmit remote voice distortion | 3.8.1.13 | 4.9 | 4.0% |
| Remote voice frequency response | 3.8.1.14 | 4.9 | 4.0% |
| Remote voice response | 3.8.2.6 | 4.9 | 4.0% |
| Remote line bias | 3.8.2.7 | 4.9 | 4.0% |
| Front end overload | 3.8.2.9 | 4.9 | 4.0% |

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IX. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection. Five (5) samples of the complete equipment shall be selected at random, one (1) sample for each Subgroup in Table IX, from the first fifty (50) or fraction thereof produced. The EMI and fungus tests shall be performed only on the first production lot. Thereafter, four (4) units of the RT-1185 shall be selected at random, one (1) sample for subgroups 1 through 4 of Table IX from every fifty (50) units or fraction of at least twenty-five (25) units produced. If the fraction is less than 25, the selection of Group C samples for the preceding test include units from the remaining fraction.



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TABLE IX. Group C inspection.

| Inspection | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Subgroup 1 | | |
| High temperature | 3.9.1.1 | 4.10.1.1 |
| Low temperature | 3.9.1.2 | 4.10.1.2 |
| Humidity | 3.9.2 | 4.10.2 |
| Subgroup 2 | | |
| Dust | 3.9.4 | 4.10.4 |
| Bounce | 3.9.5.1 | 4.10.5.1 |
| Subgroup 3 | | |
| EMI | 3.10 | 4.11 |
| Immersion | 3.9.3 | 4.10.3 |
| Shock-drop | 3.9.5.1 | 4.10.5.2 |
| Subgroup 4 | | |
| Salt fog 1/ | 3.9.6 | 4.10.6 |
| Subgroup 5 | | |
| Fungus 1/ | 3.9.7 | 4.10.7 |

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after their inspection before proceeding with subsequent inspection.

4.5.3.2 Group C failures. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the test specified in 4.14 and shall be performed on units which have been subjected to and have passed Group A and Group B inspections.

4.5.5.1 Sampling for Group D inspection. Two (2) units shall be selected at random from each successive 6 months production and subjected



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to the test of para 4.14. Samples selected from each sixth month's production may be selected early in the sixth month in order that testing be completed by the end of the sixth month.

4.5.5.2 Non-compliance. In case of Group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

4.5.5.3 Reinspection of Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.7 Burn-in. Each RT-1185 submitted for acceptance shall be subjected to burn-in in accordance with Drawing SC-A-678034. (See 3.7)

4.8 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

4.9 Performance characteristics. The performance characteristics of Receiver-Transmitter RT-1185( )/GRA-114 shall be tested in accordance with the Performance Test Procedure, Drawing SM-A-865231. Paragraph titles in the test procedure correspond to the requirement paragraphs of the specification.



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| Requirement | Paragraph |
|---|---|
| Power output | 3.8.1.1 |
| Efficiency | 3.8.1.2 |
| Encode operation | 3.8.1.3 |
| Remote encode control | 3.8.1.4 |
| Code channel frequency accuracy | 3.8.1.5 |
| Voice channel frequency accuracy | 3.8.1.6 |
| Deviation limiting | 3.8.1.7 |
| Local voice modulation capability | 3.8.1.8 |
| Local voice distortion | 3.8.1.9 |
| Local voice frequency response | 3.8.1.10 |
| Remote voice control operation | 3.8.1.11 |
| Remote voice modulation gains | 3.8.1.12 |
| Transmit remote voice distortion | 3.8.1.13 |
| Remote voice frequency response | 3.8.1.14 |
| Sensitivity | 3.8.2.1 |
| Limiting | 3.8.2.2 |
| Audio output power | 3.8.2.3 |
| Remote voice gain | 3.8.2.4 |
| Receive remote voice distortion | 3.8.2.5 |
| Remote voice response | 3.8.2.6 |
| Remote line bias | 3.8.2.7 |
| Standby input power | 3.8.2.8 |
| Front end overload | 3.8.2.9 |

4.10 Service condition. The RT-1185 shall be tested in accordance with the following conditions.

4.10.1 Temperature.

4.10.1.1 High temperature. The RT-1185 shall be subjected to the test of Method 501.1, Procedure II of MIL-STD-810. For Step 4, the chamber temperature shall be 71°C(160°F). For Step 7, the chamber temperature shall be adjusted to 65.6°C(150°F). (This includes effects of solar radiation) (See 3.9.1.1)

4.10.1.2 Low temperature. The RT-1185 shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -45.6°C(-50°F), and shall be maintained until temperature stabilization of the test item is reached. Low operating temperature (Step 4) shall be -40°C(-40°F) (see 3.9.1.2).

4.10.2 Humidity. The RT-1185 shall be subjected to the test of Method 507.1, Procedure III of MIL-STD-810. In Step 7 of Procedure III, the chassis shall be removed from its enclosure, or the test item shall be otherwise opened so as to expose to the chamber environment the normally



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sealed areas. In Step 6, measurements shall be taken during the last five hours of each test cycle. In Step 8, measurements shall be taken the last 5 hours of each 48 hours (see 3.9.2).

4.10.3 Immersion. The RT-1185 shall be subjected to the test of Method 512.1, Procedure I of MIL-STD-810 (see 3.9.3).

4.10.4 Dust. The RT-1185 shall be subjected to the test of Method 510.1, Procedure I of MIL-STD-810 (see 3.9.4).

4.10.5 Vibration, bounce and shock.

4.10.5.1 Bounce. The RT-1185 shall be subjected to the test of Method 514.2, Procedure XI, Part 2 of MIL-STD-810. It shall undergo one hour of testing at each of the following temperatures: (1) Room ambient; (2) -45°C, (-50°F); (3) 71°C, (160°F)

At each of these temperatures RT-1185 shall be vibrated for ten minutes on each face (see 3.9.5.1).

4.10.5.2 Shock, drop. The RT-1185 shall be subjected to the test of Method 516.2, Procedure II of MIL-STD-810 (see 3.9.5.1).

4.10.6 Salt fog. The RT-1185 shall be subjected to the test of Method 509.1, Procedure I of MIL-STD-810 (see 3.9.6).

4.10.7 Fungus. The RT-1185 shall be subjected to the test of Method 508.1, Procedure I of MIL-STD-810. Gasket sealed or nonhermetic sealed test items shall be opened during exposure and internal areas, as well as external surfaces, shall be sprayed with spore suspension. Operation of the equipment is not required (see 3.9.7).

4.11 Electromagnetic interference characteristics. The RT-1185 shall be tested in accordance with MIL-STD-462 and meet requirements of 3.10.

4.12 Weight. The RT-1185 shall be weighed and shall meet the requirement of 3.11.

4.13 Visual and mechanical inspection. The RT-1185 shall be examined for the applicable defects listed in MIL-STD-252.

4.14 Reliability testing.

4.14.1 First article and quality conformance Group D inspection. To determine compliance with paragraph 3.12 during First Article and Quality Conformance inspection, the equipment shall be tested in accordance with Test Plan XXV of MIL-STD-781. The specified MTBF for all reliability testing shall be 1040 hours.



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4.14.2 Failure definition. A failure is defined as the inability of the equipment to perform its required functions as specified in this specification and in the contractor prepared, government approved reliability test plan.

4.14.3 Test conditions. The test shall be conducted under the conditions of Test Level C of MIL-STD-781 with the exception of no vibration. The test shall be conducted in accordance with the reliability test plan required in the contract.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.(2)(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Receiver-Transmitter RT-1185( )/GRA-114 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Receiver-Transmitter RT-1185( )/GRA-114 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Receiver-Transmitter RT-1185( )/GRA-114. Each RT-1185 shall be individually unit packed method III as follows: Place the required components within the space provided and secure the fasteners. Cushion each unit on all surfaces with cells or pads, or both, fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the container. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservation application and unit packaging. Each unit shall be cleaned, dried, preserved and unit packed as specified in 5.1.1.



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5.1.3 Commercial. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.(2)(b)). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Receiver-Transmitter RT-1185( )/GRA-114, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.2 Level B. A quantity of Receiver-Transmitter RT-1185( )/GRA-114, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be specified in the appendix of the specification. A quantity of the containers shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-640, shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers shall be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.



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5.2.2.1 Unitization. Palletization shall be required when: containers specified in 5.2.2 do not require pallet-mounting; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.2, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.3 Commerical packing. Commercial packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. The Receiver-Transmitter RT-1185( )/GRA-114 is used in conjunction with Receiver-Transmitter RT-1183( )/GRA-114 and Receiver-Transmitter RT-1184( )/GRA-114 and is intended to provide a system control and a voice and data transmission facility as part of Radio Data Link Sound Ranging AN/GRA-114.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level A, level B or commercial preservation and packing (see section 5).

c. When rough handling test is required.

d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

e. Marking and shipping of samples.

f. Place of final inspection.



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g. Number of first articles to be submitted for approval. If other than called for in specification (see 3.1).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted and replaced by a letter identifying the particular design for example, RT-1185W/GRA-114. The contractor should apply for nomenclature in accordance with applicable clause in the contract (see 1.1).

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Preparing Activity:
Army-EL
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DOCUMENT IDENTIFIER AND TITLE MIL-C-49113(EL) RECEIVER-TRANSMITTER RT-1185( )/
GRA-114

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
|---|---|
| | MATERIAL PROCURED UNDER A<br>☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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